

# Hardware Project No. 4

## Bridge Circuit for Transducer Interface

ECE 444: Bioinstrumentation and Sensing

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### 1. Introduction

Bridge circuits are the standard front end for resistive transducers in biomedical and physical instrumentation. A resistive transducer such as a strain gauge changes resistance in response to some physical stimulus, but the resistance change is normally a small fraction of the gauge's resting value. Measuring such a small fractional resistance change directly is awkward; placing the gauge in one arm of a Wheatstone bridge converts the fractional resistance change into a small differential voltage that scales linearly with the change for small inputs. With a stable excitation source and downstream amplification, this differential voltage can then be measured cleanly.

This report describes the construction and testing of a strain-gauge Wheatstone bridge for the ECE 444 transducer-interface project. The bridge was excited by a regulated 5 V supply derived from an LM7805 three-terminal voltage regulator. The intended signal chain was bridge, then three-op-amp instrumentation amplifier (re-using the LF411 circuit from Hardware Project No. 1), then oscilloscope. The bridge stage itself was successful: with no amplification, the raw bridge output shifted by a few millivolts in response to manually applied strain, consistent with the expected quarter-bridge transfer characteristic. The amplifier-integration stage, however, was not successful. When the bridge output was wired into the instrumentation amplifier, the amplifier output saturated near the negative supply rail and the strain gauge subsequently overheated and was destroyed before a usable amplified measurement could be recorded.

What follows is a complete account of the working bridge stage, the failed amplifier integration, and the most plausible explanations for why the integration failed. The bridge results are presented as measured; the amplifier integration is treated as a failure analysis rather than a successful measurement.

### 2. Theory and Design Derivation

#### 2.1 Wheatstone Bridge with Resistive Transducer

The Wheatstone bridge consists of two voltage dividers in parallel, both fed from the same excitation source. The differential output is taken between the two midpoint nodes. The arrangement used in this project places the strain gauge in arm  $R_1$  and three nominally equal fixed resistors  $R$  in the remaining three arms, as shown in Figure 1. Writing the strain-gauge resistance as  $R(1 + X)$ , where  $X$  is the fractional resistance change  $\Delta R/R$ , the open-circuit bridge output follows directly from two voltage-divider expressions:

$$E_0 = [R/(1 + X) / (R + R/(1 + X))] \cdot V_{IN} - (1/2) \cdot V_{IN}.$$

Combining the two terms over a common denominator and simplifying gives the closed-form result

$$E0 = (VIN / 4) \cdot X / (1 + X/2),$$

which in the small-strain limit ( $|X| \ll 1$ ) reduces to the well-known quarter-bridge approximation

$$E0 \approx (VIN / 4) \cdot X.$$

This is the expression printed alongside the bridge schematic in Figure 1. Two features of this result drove the design decisions for the rest of the circuit. First, the output is linear in  $X$  to first order, with a sensitivity of  $VIN / 4$ . With  $VIN = 5$  V, the sensitivity is 1.25 V per unit of fractional resistance change, so a 0.1% change in the gauge resistance produces a bridge output of 1.25 mV. Second, the output is proportional to the excitation voltage, so any drift or noise on  $VIN$  appears directly at the bridge output. A regulated, stable excitation source is therefore essential.

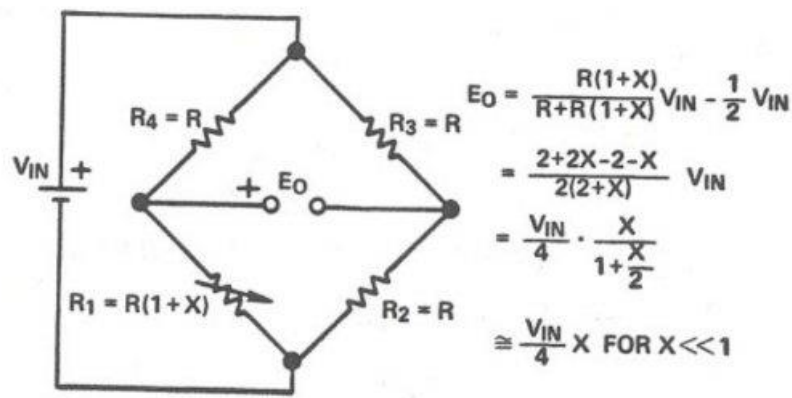


Figure 1. Wheatstone bridge circuit with a resistive transducer in arm  $R1$ , reproduced from the assignment handout. The strain gauge has resistance  $R(1 + X)$ , where  $X = \Delta R / R$  is the fractional resistance change. The remaining three arms are nominally equal fixed resistors  $R$ . The output voltage  $E0$  is taken differentially between the two bridge midpoint nodes; the derivation alongside the schematic shows that for  $X \ll 1$  the output reduces to  $E0 \approx (VIN / 4) \cdot X$ , the quarter-bridge approximation used in this project.

## 2.2 Strain Gauge Operating Principle

A strain gauge is a thin resistive grid bonded to a flexible carrier; when the carrier stretches or compresses, the grid deforms with it and its electrical resistance changes by a small amount. The defining figure of merit is the gauge factor, defined as the ratio of the fractional resistance change to the mechanical strain:

$$GF = (\Delta R / R0) / (\Delta L / L0).$$

The handout specifies a nominal gauge resistance of  $120 \pm 3 \Omega$ , a gauge factor of  $2.1 \pm 1\%$ , and a strain limit of 2%. The handout also explicitly invites the user to verify the unloaded gauge resistance with a meter, which proved important here: the gauge actually used measured approximately

$$Rg \approx 532 \Omega,$$

which is roughly four times the nominal  $120 \Omega$  value. Whether this is a labeling error, an out-of-tolerance sample, or simply a different gauge from the one specified is not certain, but the practical consequence is unambiguous: the three fixed bridge arms must be chosen to match the measured gauge resistance, not the

handout's nominal value, because a Wheatstone bridge balances only when the resistor ratios on its two legs are matched. The fixed resistors were therefore chosen to be approximately  $532\ \Omega$  rather than  $120\ \Omega$ .

## 2.3 Excitation Voltage Regulation

Because the bridge output is proportional to the excitation voltage, the excitation must be stable on the same timescale as the measurement. Bench supplies can have surface-level noise on the order of millivolts, which is comparable to the bridge output itself; a dedicated three-terminal linear regulator removes this problem at essentially zero cost in circuit complexity. The LM7805 supplied with the project kit was used as the bridge-excitation regulator. The datasheet recommends an input voltage range of

$$7\text{ V} \leq V_I \leq 25\text{ V}$$

and specifies a nominal output of 5 V. A bench supply set to roughly 9 V to 12 V was used as the regulator input. Per the datasheet's standard fixed-regulator application, a  $0.33\ \mu\text{F}$  capacitor was placed across the input and a  $0.1\ \mu\text{F}$  capacitor across the output to provide local high-frequency bypass. The regulator's common terminal served as the ground node for the bridge.

## 2.4 Amplification of the Bridge Output

A quarter-bridge sensitivity of  $V_{IN} / 4 = 1.25\text{ V}$  per unit  $X$  means that even a 0.1% fractional resistance change produces only 1.25 mV at the bridge output. The mechanical strain achievable by hand pressure on a small gauge is well below 1% strain (the gauge's strain limit), and the multiplicative effect of the gauge factor ( $GF \approx 2.1$ ) implies fractional resistance changes of the order  $10^{-3}$  at best. The expected bridge output is therefore at most a few millivolts, in line with what the handout itself anticipates when it remarks that millivolt outputs would call for amplification. Direct measurement of millivolt-level signals on a bench oscilloscope is possible but inconvenient, so the bridge was intended to drive a discrete three-op-amp instrumentation amplifier built earlier for Hardware Project No. 1.

Figure 2 shows the conceptual topology: the bridge sits on the left, and its two midpoint nodes drive the differential inputs of an instrumentation amplifier whose differential gain is set by a single internal (or in this case external, hand-built) resistor. For a discrete three-op-amp in-amp with first-stage feedback resistors  $R_2$  and gain-setting resistor  $R_1$ , the closed-form differential gain derived in the Project No. 1 report is

$$G = 1 + 2 R_2 / R_1.$$

The previously constructed in-amp used  $R_1 = 400\ \Omega$  (two  $200\ \Omega$  resistors in series) and  $R_2 = 100\text{ k}\Omega$ , giving a nominal differential gain of

$$G = 1 + 2(100\ 000) / 400 = 501.$$

The measured gain at 100 Hz, again from the Project No. 1 report, was 520 V/V, in agreement with the nominal value to within ordinary 5%-resistor tolerance. This is a very high gain for a transducer interface and, as discussed in Section 6, was almost certainly the proximate cause of the eventual amplifier saturation in this project.

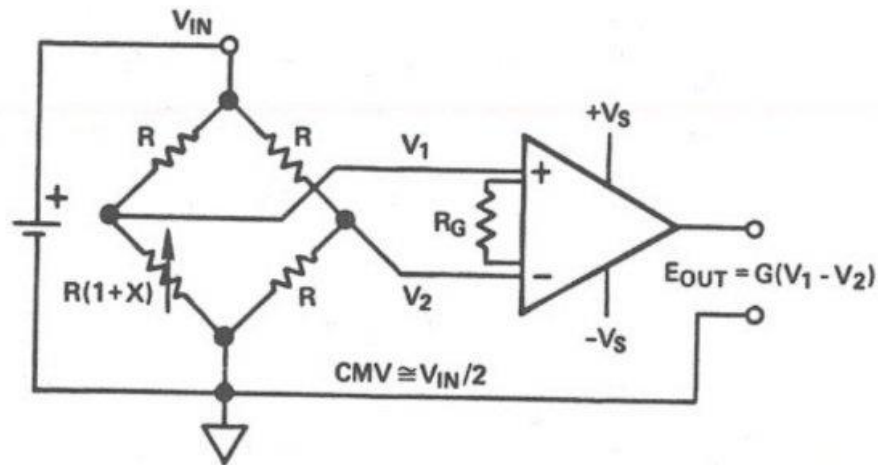


Figure 2. Instrumentation-amplifier-based bridge amplifier topology, reproduced from the assignment handout (Figure 2b). The bridge on the left drives the differential inputs  $V_1$  and  $V_2$  of an instrumentation amplifier whose differential gain is set by a single gain resistor  $R_G$ . The amplifier output is  $E_{OUT} = G(V_1 - V_2)$ . The common-mode voltage at the in-amp inputs sits at approximately  $V_{IN}/2$ , which is well within the input common-mode range of any reasonable op amp powered from a dual supply. The crucial advantage of this configuration over the single-op-amp bridge amplifier is that the in-amp's high input impedance does not load the bridge.

### 3. Circuit Construction

#### 3.1 Voltage Regulator Stage

The LM7805 was wired in its standard fixed-output configuration. The bench supply positive output was connected to the regulator's input pin, the bench supply ground was connected to the regulator's common pin, and the regulator's output pin became the +5 V bridge-excitation node. A 0.33  $\mu\text{F}$  capacitor bridged the input pin to common, and a 0.1  $\mu\text{F}$  capacitor bridged the output pin to common, both placed physically close to the regulator package. The regulator output was verified with a multimeter to be within a few tens of millivolts of 5 V before any bridge load was applied.

#### 3.2 Bridge Stage

The bridge was assembled on a solderless breadboard. The strain gauge occupied one arm. The remaining three arms each consisted of resistors chosen to approximately match the measured strain-gauge resistance of 532  $\Omega$ . The midpoint nodes  $V_1$  and  $V_2$  were each brought out to dedicated breadboard rails so they could be probed individually with a multimeter or, later, wired into the in-amp inputs. With the bridge powered but no strain applied, each midpoint was expected to sit at approximately

$$V_1 \approx V_2 \approx V_{IN} / 2 = 2.5 \text{ V},$$

and the differential bridge output  $E_0 = V_1 - V_2$  was expected to be small (ideally zero, in practice a few millivolts of offset due to the unavoidable mismatch among 5% resistors and the gauge).

#### 3.3 Instrumentation Amplifier Stage

The discrete LF411 three-op-amp instrumentation amplifier from Hardware Project No. 1 was carried over unchanged. The first-stage amplifiers used  $R_1 = 400 \Omega$  and  $R_2 = 100 \text{ k}\Omega$ , and the second-stage difference amplifier used four equal 10  $\text{k}\Omega$  resistors. The amplifier was powered from a dual  $\pm 12 \text{ V}$  bench

supply, with pin 7 of each LF411 to +12 V, pin 4 to −12 V, pin 3 the non-inverting input, pin 2 the inverting input, and pin 6 the output. The bridge ground (which is the LM7805 common) and the dual-supply ground reference were tied together at a single point so the amplifier and the bridge would share a common 0 V reference. The two bridge midpoint nodes were intended to drive the two amplifier inputs through short jumper wires; in the actual integration these connections produced the failure documented in Section 5.2.

| Component                     | Nominal Value                    | Function                                                      |
|-------------------------------|----------------------------------|---------------------------------------------------------------|
| Strain gauge                  | $\approx 532\ \Omega$ (measured) | Resistive transducer; one arm of the bridge                   |
| R ( $\times 3$ , bridge arms) | $\approx 532\ \Omega$            | Three fixed bridge arms chosen to match the gauge             |
| LM7805                        | 5 V regulated output             | Bridge excitation supply                                      |
| C in                          | 0.33 $\mu\text{F}$               | Regulator input bypass capacitor                              |
| C out                         | 0.1 $\mu\text{F}$                | Regulator output bypass capacitor                             |
| R1 (in-amp)                   | 400 $\Omega$                     | First-stage gain-setting resistor (sets $G = 1 + 2 R2 / R1$ ) |
| R2 ( $\times 2$ , in-amp)     | 100 k $\Omega$                   | First-stage feedback resistors                                |
| R ( $\times 4$ , in-amp)      | 10 k $\Omega$                    | Difference-amplifier resistors                                |
| A1, A2, A3                    | LF411                            | JFET-input op amps for the three in-amp stages                |
| In-amp supplies               | $\pm 12\ \text{V}$ dual rail     | Bench dual supply                                             |

*Table 1. Component summary for the bridge-with-amplifier signal chain. The in-amp values are reproduced unchanged from the Hardware Project No. 1 build.*

## 4. Experimental Procedure

The experiment was carried out in five steps, intentionally staged so that each subsystem could be verified before the next was connected.

First, the LM7805 was wired with its input and output bypass capacitors and powered from a bench supply at approximately 9 V. The regulator output was measured with a multimeter against the regulator common pin and confirmed to be within tolerance of 5 V before any bridge load was applied.

Second, the strain-gauge resistance was measured with a multimeter against the gauge leads. This was the step at which the gauge was found to read approximately 532  $\Omega$  rather than the handout-specified 120  $\Omega$ . The three fixed bridge resistors were then chosen to approximately match the measured 532  $\Omega$  value, and the bridge was assembled on the breadboard.

Third, with the bridge powered and no strain applied, the two midpoint voltages V1 and V2 were each measured against ground, and the differential bridge output  $E0 = V1 - V2$  was measured directly between the two midpoint nodes.

Fourth, mechanical strain was applied manually to the gauge while the multimeter remained connected across the bridge midpoints. The output was observed while strain was applied and released, and the change in  $E_0$  was recorded. At this point the bridge had not yet been connected to any amplifier, so all measurements were of the raw, unamplified bridge output.

Fifth, the two bridge midpoint nodes were wired into the inputs of the previously built LF411 instrumentation amplifier, the in-amp dual  $\pm 12$  V supply was switched on, and the amplifier output was probed. It was during this final step that the failure described in Section 5.2 occurred.

## 5. Results

### 5.1 Raw Bridge Output

Before the amplifier was connected, the bridge produced the small, repeatable differential output expected of a working strain-gauge bridge. With no strain applied, the bridge sat near balance with a small residual offset:

$$E_0, \text{ no strain} \approx 1.76 \text{ mV}.$$

When manual strain was applied to the gauge, the differential output shifted consistently to

$$E_0, \text{ strained} \approx -0.4 \text{ mV},$$

for a measured change of

$$\Delta E_0 = E_0, \text{ strained} - E_0, \text{ no strain} \approx -2.16 \text{ mV}.$$

The sign of the change depends on which midpoint was wired to the meter's positive lead and whether the gauge was stretched or compressed in the applied strain; the magnitude of the change, on the order of a few millivolts, is the result of interest. Releasing the strain returned the output close to the original no-strain offset of 1.76 mV, confirming that the bridge was responding reversibly to the applied mechanical input rather than drifting.

| Condition               | Raw bridge output $E_0$ |
|-------------------------|-------------------------|
| No applied strain       | 1.76 mV                 |
| Applied strain (manual) | -0.4 mV                 |
| Change $\Delta E_0$     | -2.16 mV                |

*Table 2. Measured raw bridge output before any amplification. Values were read directly from a digital multimeter connected differentially between the two bridge midpoint nodes.*

The 1.76 mV no-strain offset is consistent with the unavoidable resistor mismatch among the bridge arms. Using the small-signal sensitivity  $E_0 \approx (V_{IN} / 4) \cdot X$  with  $V_{IN} = 5$  V, the offset corresponds to an effective fractional resistance imbalance of

$$X \approx 4 \cdot (1.76 \text{ mV}) / (5 \text{ V}) \approx 1.4 \times 10^{-3},$$

or about 0.14%. This is well inside the 5% resistor tolerance budget and is the kind of imbalance that an in-amp's offset-trim potentiometer is normally used to null out before measurement. The strain-induced change  $\Delta E_0 \approx -2.16$  mV corresponds to an effective fractional resistance change of approximately 0.17% in the gauge, consistent with a modest hand-applied strain through a gauge factor near 2.

## 5.2 Amplifier Integration Result

Connecting the bridge to the instrumentation amplifier produced an immediate failure of the integrated system. With the in-amp's dual  $\pm 12$  V supplies switched on, the amplifier output rail came up close to the negative supply rail and stayed there:

$$V_{out} \approx -12 \text{ V}.$$

This is rail saturation, not an amplified signal. The amplifier had moved out of its linear operating region and was no longer functioning as a small-signal amplifier in any useful sense. No amplified strain measurement could be recorded under this condition.

Shortly after the integrated circuit was powered, the strain gauge itself overheated visibly and was thermally destroyed, with its bonded element melting away from the carrier. Once the gauge was destroyed it could not be brought back online, and no further amplified measurements were possible with this gauge.

| Test                           | Outcome                                               |
|--------------------------------|-------------------------------------------------------|
| Bridge alone                   | Repeatable millivolt-level response to applied strain |
| In-amp connected to bridge     | Amplifier output saturated near $-12$ V               |
| Strain gauge after integration | Overheated and physically destroyed                   |
| Amplified strain measurement   | Not obtained                                          |

*Table 3. Summary of integrated-system test outcomes. The bridge stage worked in isolation; the amplifier-integration stage did not.*

## 6. Discussion

### 6.1 Bridge Stage: What Worked

The bridge stage operated as predicted by the quarter-bridge model in Section 2.1. A repeatable, small differential output appeared at rest, shifted by a measurable amount in response to applied strain, and returned to the rest value when the strain was released. The magnitude of the shift, on the order of 2 mV, is exactly the millivolt-level signal that the handout warned about when it remarked that bridge transducers usually need amplification. The bridge therefore functioned correctly as a transducer interface; the difficulty in this project lay entirely downstream of the bridge.

### 6.2 Why the High-Gain In-Amp Was the Wrong Choice

The amplifier carried over from Hardware Project No. 1 had a measured differential gain of approximately 520 V/V. With  $\pm 12$  V supplies, the LF411 output can swing to within about a volt of either rail, so the maximum useful differential input before saturation is roughly

$$V_{diff, sat} \approx 11 \text{ V} / 520 \approx 21 \text{ mV}.$$

Even ignoring offsets, this is dangerously close to the operating range of the bridge itself: the measured no-strain offset of 1.76 mV is already 8% of the way to saturation, and any additional input offset from the LF411 (typically a few millivolts referred to the input, though usually much smaller for a JFET-input device) gets multiplied by the same factor of 520. With the in-amp's input-offset trim left in whatever

position it was in at the end of the Project No. 1 measurements, it is very plausible that the combined offset of the bridge plus the in-amp's own input offset, both multiplied by 520, was already past the saturation threshold before any strain was even applied. The honest interpretation is that a gain of 500 was simply too aggressive for this application; a gain of order 50 to 100 would have left far more headroom for the unavoidable bridge offset and would not have been close to saturation.

### 6.3 Other Plausible Contributors to the Saturation

Three additional explanations cannot be ruled out from the available evidence, and the actual cause may be some combination of them with the high-gain issue above.

First, a grounding error. The bridge ground (the LM7805 common) and the in-amp's 0 V supply reference have to be physically tied together for the in-amp to see meaningful single-ended voltages at its inputs. If that common-tie was missing or had a bad breadboard contact, the two bridge midpoints would float at some indeterminate DC level relative to the in-amp's supplies, and the amplifier would have no reason to sit anywhere except at one of its rails.

Second, a wiring error in the difference-amplifier stage. The Project No. 1 report documents that an initial transposition of the difference-amplifier input wires turned the output into a positive-feedback latch that sat at one of the rails until corrected. If the difference-amp wiring was disturbed between Project No. 1 and the present project (which is easy to do on a breadboard), the same failure mode would reappear.

Third, an input-bias-current return path issue. The LF411 inputs are JFETs and need only picoamps of bias current, but they still need a DC path to the in-amp's reference ground. If the bridge midpoints were not DC-coupled to ground through the bridge (they are, through the lower bridge arms, so this should not be the issue here), the in-amp inputs would drift. This is unlikely to be the dominant cause in this circuit, but it is the kind of subtle integration error that is worth mentioning for completeness.

### 6.4 Strain Gauge Overheating

The strain-gauge destruction is the most concerning observation, because in a correctly wired bridge the gauge dissipates very little power. With  $V_{IN} = 5\text{ V}$  across the two series-connected  $532\ \Omega$  arms of one bridge leg, the branch current is

$$I_{\text{branch}} = 5\text{ V} / (532\ \Omega + 532\ \Omega) \approx 4.70\text{ mA},$$

and the power dissipated in the gauge itself is

$$P = I^2 R = (4.70\text{ mA})^2 \cdot 532\ \Omega \approx 11.8\text{ mW},$$

which is well within the thermal capacity of even a small foil gauge. A gauge does not melt at 12 mW. The fact that this gauge did overheat and melt therefore strongly implies that, during the integration stage, current was reaching the gauge through some path other than the intended bridge leg. The most plausible scenario is an inadvertent contact between one of the in-amp input wires and the regulated 5 V rail (or, worse, one of the  $\pm 12\text{ V}$  supply rails), which would have placed a far larger voltage across the gauge than the bridge ever does on its own. A short from one of the in-amp supplies directly across the gauge would dump enough power to melt the bond. This kind of accidental contact is a constant hazard when adding wires to a working breadboard, and it cannot be ruled out from what was observed.

### 6.5 What a Second Attempt Should Do Differently

Several corrective measures, in roughly decreasing order of expected effect, would make a future attempt much more likely to succeed.

First, lower the in-amp differential gain to something compatible with a bridge front end. Keeping  $R_2 = 100\text{ k}\Omega$  and switching  $R_1$  from  $400\text{ }\Omega$  to  $2\text{ k}\Omega$  gives

$$G = 1 + 2(100\,000) / 2000 = 101,$$

which would produce an output change of about  $0.22\text{ V}$  for the measured  $2.16\text{ mV}$  strain response while still leaving over a hundred millivolts of input headroom before saturation. A gain of order 100 is the right ballpark for this kind of bridge interface.

Second, verify the in-amp in isolation before connecting the bridge. Short the two amplifier inputs together (so the differential input is zero by construction) and confirm that the output sits within a small fraction of a volt of ground with both supplies on. If the output goes to a rail under this test, the amplifier itself is miswired and connecting the bridge would only have masked the problem.

Third, confirm a shared ground reference between the LM7805 common and the in-amp's  $\pm 12\text{ V}$  supply ground at a single, physically robust tie point. A separate piece of wire dedicated to this tie, with both ends in known-good breadboard rows, eliminates the most common integration failure mode.

Fourth, install a small series resistor (say  $1\text{ k}\Omega$ ) on each of the wires running from the bridge midpoints into the in-amp inputs. This adds a negligible voltage error to the measurement (in-amp input bias currents are picoamps) but converts any accidental short between an in-amp input wire and a supply rail from a destructive event into a survivable one. This protection alone would very likely have saved the gauge.

Fifth, use the in-amp's input-offset trim potentiometer to null the bridge no-strain offset before applying strain. With a properly set null and a gain of 100, the system would be measuring small changes around zero output rather than small changes around a large amplified offset.

## 7. Error Sources

The dominant error sources, listed in roughly decreasing order of effect on the measurements actually obtained, are as follows.

**Excessive amplifier gain.** The carried-over  $G = 520$  in-amp was simply too high for a bridge front end and is the most likely proximate cause of the saturation failure. This is a design choice rather than a tolerance issue, and it is correctable at the cost of two resistors.

**Integration grounding or wiring error.** The amplifier worked correctly in Hardware Project No. 1 and the bridge worked correctly alone in this project, but the integrated system did not. A subtle grounding or wiring mistake at the integration interface is therefore implicated by elimination, even though the specific mistake was not isolated before the gauge was destroyed.

**Strain-gauge thermal destruction.** Once the gauge melted, no further measurements were possible with that sensor. The most plausible mechanism is an accidental short from an in-amp supply rail across the gauge during the integration step, since the bridge's own  $12\text{ mW}$  of nominal gauge dissipation is far below any reasonable failure threshold.

**Bridge-arm mismatch.** The fixed bridge resistors and the strain gauge are all 5%-tolerance parts; the measured no-strain offset of 1.76 mV corresponds to a roughly 0.14% imbalance, which is well within the expected tolerance budget but consumes input headroom that, at a gain of 520, is no longer negligible.

**Manual strain application.** Strain was applied by hand, so the input was neither calibrated nor reproducible. This limits the bridge-stage results to qualitative confirmation that the gauge responded to strain, not to quantitative gauge-factor extraction.

**Multimeter resolution.** Measuring a 2 mV differential change with a handheld multimeter introduces a non-negligible quantization and noise floor; an oscilloscope on a millivolt-per-division setting, or an amplified version of the bridge output, would have been more appropriate for fine measurement and is in fact the entire reason amplification was being added in the first place.

## 8. Conclusion

A Wheatstone bridge with a strain gauge in one arm was constructed and tested as a transducer interface. The bridge was excited by a regulated 5 V supply from an LM7805 three-terminal regulator and was matched to the measured 532  $\Omega$  strain-gauge resistance rather than the handout's nominal 120  $\Omega$ . With no amplification, the bridge produced a small but repeatable differential output that shifted from approximately 1.76 mV at rest to approximately -0.4 mV under manually applied strain, a change of about 2.16 mV. This is consistent with the quarter-bridge approximation  $E_0 \approx (V_{IN} / 4) \cdot X$  and corresponds to a fractional resistance change in the gauge of approximately 0.17%.

The downstream integration with the three-op-amp LF411 instrumentation amplifier from Hardware Project No. 1 was unsuccessful. When the bridge midpoints were wired to the amplifier inputs, the amplifier output saturated near the -12 V supply rail and the strain gauge subsequently overheated and was destroyed. The most plausible explanation is the combination of an excessive in-amp differential gain ( $G \approx 520$ , which leaves only about 21 mV of differential input headroom before saturation) and an integration-stage wiring or grounding error that was never isolated because the destruction of the gauge ended the experiment. The gauge melting in particular strongly suggests an accidental short from an in-amp supply rail to the gauge during integration; nominal bridge currents are nowhere near enough to thermally destroy a foil gauge on their own.

The bridge stage of the project therefore met its objective in isolation: a strain-gauge Wheatstone bridge does convert a small fractional resistance change into a measurable differential voltage at the expected sensitivity. The amplifier stage did not. A second attempt, with a much lower in-amp gain ( $G \approx 100$ ), in-isolation verification of the amplifier with shorted inputs before bridge connection, an explicitly verified shared ground tie between the LM7805 common and the  $\pm 12$  V supply ground, and small series-protection resistors on the in-amp input wires, would address the failure modes identified here and is the recommended path forward.

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